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PUBLICATIONS & PRE-PRINTS

Total Publications (15)

Total Citations: 464

First-authored Publications (6)

Total Citations: 171

- [1] **Z. C. Zeng**, A. Peter, L. Strigari, A. Benson, X. Du, C. Mace, *Feeding the Void: Co-evolution of the SIDM-seeded Black Hole and the dark halo after core-collapse*, 2026, arXiv:2609.22758. cited by: 0
- [2] **Z. C. Zeng**, O. Hartl, L. Strigari, A. Peter, X. Du, C. Mace, A. Benson, *Bypassed Core Formation in Milky Way-Mass SIDM Halos: Implications for the Local Group Past-Pericenter Scenario*, 2026, arXiv:2604.08647. cited by: 2
submitted to Phys.Rev.D
- [3] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, J. Li, et al., *Diversity and universality: evolution of dwarf galaxies with self-interacting dark matter*, 2025, Phys.Rev.D, arXiv:2412.14621. cited by: 14
- [4] **Z. C. Zeng**, A. Peter, X. Du, S. Yang, A. Benson, et al., *Till the core collapses: the evolution and properties of self-interacting dark matter subhalos*, 2025, Phys.Rev.D, arXiv:2310.09910. cited by: 30
- [5] **Z. C. Zeng**, A. Peter, X. Du, A. Benson, S. Kim, et al., *Core-collapse, evaporation and tidal effects: the life story of a self-interacting dark matter subhalo*, 2022, MNRAS, arXiv:2110.00259. cited by: 105
- [6] **Z. C. Zeng**, S. Yeung and M.-C. Chu, *Effects of neutrino mass and asymmetry on cosmological structure formation*, 2019, JCAP, arXiv:1808.00357. cited by: 20

Mentored Student Publications (3)

Total Citations: 60

- [1] C. Mace, B. Dhanasingham, **Z. C. Zeng**, et al., *The Sensitivity of Substructure Lensing to SIDM Core-collapse Model Variation*, 2026, Phys.Rev.D, arXiv:2605.24174. cited by: 3
- [2] C. Mace, S. Yang, **Z. C. Zeng**, et al., *Calibrating the SIDM Gravothermal Catastrophe with N-body Simulations*, 2026, Phys.Rev.D, arXiv:2504.13004. cited by: 14
- [3] C. Mace, **Z. C. Zeng**, A. Peter, et al., *Convergence Tests of Self-Interacting Dark Matter Simulations*, 2024, Phys.Rev.D, arXiv:2402.01604. cited by: 43

Collaborative Publications (6)

Total Citations: 233

- [1] A. Engelhardt, et al. (inc. **Z. C. Zeng**), *MARVELously Dark: the density profile evolution of dwarf halos in velocity-dependent SIDM*, 2026, JCAP, arXiv:2601.23264. cited by: 8
- [2] X. Du, A. Benson, **Z. C. Zeng**, et al., *Tidal evolution of cored and cuspy dark matter halos*, 2024, Phys.Rev.D, arXiv:2403.09597. cited by: 34
- [3] S. Yang, et al. (inc. **Z. C. Zeng**), *A quantitative comparison between velocity dependent SIDM cross-sections constrained by the gravothermal and isothermal models*, 2024, MNRAS, arXiv:2305.05067. cited by: 28
- [4] S. Yang, X. Du, **Z. C. Zeng**, A. Benson, F. Jiang, E. O. Nadler, A. Peter, *Gravothermal Solutions of SIDM Halos: Mapping from Constant to Velocity-dependent Cross Section*, 2023, ApJ, arXiv:2205.02957. cited by: 82
- [5] F. Jiang, et al. (inc. **Z. C. Zeng**), *A semi-analytic study of self-interacting dark-matter haloes with baryons*, 2023, MNRAS, arXiv:2206.12425. cited by: 63
- [6] R. Emami, et al. (inc. **Z. C. Zeng**), *Evidence of Neutrino Enhanced Clustering in a Complete Sample of Sloan Survey Clusters, Implying $\sum m_\nu = 0.119 \pm 0.034$ eV*, 2017, arXiv:1711.05210. cited by: 18